

Assignment: 5

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Q. (1) we have:

$$ds^2 = -dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$$

Metric for spherical polar co-ordinates with sign $(-, +, +)$ is diagonal:

$$g_{\mu\nu} = \text{diag}(-1, 1, r^2, r^2 \sin^2 \theta)$$

$$\text{So, } \begin{array}{l|l} g^{tt} = -1 & g^{\theta\theta} = \frac{1}{r^2} \\ g^{rr} = 1 & g^{\phi\phi} = \frac{1}{r^2 \sin^2 \theta} \end{array}$$

$\Rightarrow$ coordinates: $x^0 = t$, $x^1 = r$, $x^2 = \theta$, $x^3 = \phi$

The Christoffel symbols are:

$$\Gamma_{\mu\nu}^{\lambda} = \frac{1}{2} g^{\lambda\sigma} (\partial_{\mu} g_{\nu\sigma} + \partial_{\nu} g_{\mu\sigma} - \partial_{\sigma} g_{\mu\nu})$$

$\therefore$ only off-diagonal metric derivatives are non-zero:

$$\partial_r g_{\theta\theta} = \frac{d}{dr}(r^2) = 2r, \quad \partial_r g_{\phi\phi} = \frac{d}{dr}(r^2 \sin^2 \theta) = 2r \sin^2 \theta$$

$$\partial_{\theta} g_{\phi\phi} = \frac{d}{d\theta}(r^2 \sin^2 \theta) = 2r^2 \sin \theta \cos \theta$$

$$\text{All } \partial_t = 0$$

And all ∂_t and ∂_{ϕ} terms vanish.

So; the non-zero symbols (symmetric in lower indices):

$\Rightarrow$ for $\lambda = r$ (from $g^{rr} = 1$):

$$\Gamma_{\theta\theta}^r = \frac{1}{2} (-\partial_r g_{\theta\theta}) = \frac{1}{2} (-2r) = -r$$

$$\Gamma_{\phi\phi}^r = \frac{1}{2} (-\partial_r g_{\phi\phi}) = \frac{1}{2} (-2r \sin^2 \theta) = -r \sin^2 \theta$$

$\Rightarrow$ for $\lambda = \theta$ (from $g^{\theta\theta} = 1/r^2$):

$$\Gamma_{r\theta}^{\theta} = \Gamma_{\theta r}^{\theta} = \frac{1}{2} g^{\theta\theta} \partial_r g_{\theta\theta} = \frac{1}{2r^2} (2r) = \frac{1}{r}$$

$$\Gamma_{\phi\phi}^{\theta} = \frac{1}{2} g^{\theta\theta} (-\partial_{\theta} g_{\phi\phi}) = \frac{1}{2r^2} (-2r^2 \sin \theta \cos \theta)$$

$$= -\sin \theta \cos \theta$$

$\Rightarrow$ for $\lambda = \phi$ (from $g^{\phi\phi} = 1/r^2 \sin^2 \theta$):

$$\Gamma_{r\phi}^{\phi} = \Gamma_{\phi r}^{\phi} = \frac{1}{2} g^{\phi\phi} \partial_r g_{\phi\phi} = \frac{1}{2r^2 \sin^2 \theta} (2r \sin^2 \theta)$$

$$\Gamma_{\theta\phi}^{\phi} = \Gamma_{\phi\theta}^{\phi} = \frac{1}{2} g^{\phi\phi} \partial_{\theta} g_{\phi\phi} = \frac{1}{r}$$

$$= \frac{1}{2r^2 \sin^2 \theta} (2r^2 \sin \theta \cos \theta) = \cot \theta$$

All other components vanishes due to no relevant derivatives or metric components.

Q(2). The Schwarzschild metric is $ds^2 = -\alpha dt^2 + \alpha^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$, where

$$\alpha = 1 - 2GM/r$$

Thus, $g_{tt} = -\alpha$, $g_{rr} = \alpha^{-1}$, $g_{\theta\theta} = r^2$, $g_{\phi\phi} = r^2 \sin^2 \theta$

& $g^{tt} = -\alpha^{-1}$, $g^{rr} = \alpha$, $g^{\theta\theta} = 1/r^2$, $g^{\phi\phi} = 1/(r^2 \sin^2 \theta)$

so; $\partial_r \alpha = \partial_r (1 - 2GM/r) = \frac{2GM}{r^2}$

$$\Rightarrow \partial_r g_{tt} = -\partial_r \alpha = -\frac{2GM}{r^2}$$

$$\partial_r g_{rr} = \partial_r (\alpha^{-1}) = -\alpha^{-2} \partial_r \alpha = -\frac{2GM}{r^2 \alpha^2}$$

& angular part matches with Question (1).

Now, using: $\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$

$$\Rightarrow \Gamma_{tr}^t = \Gamma_{rt}^t = \frac{1}{2} g^{tt} \partial_r g_{tt} = \frac{1}{2} (-\alpha^{-1}) \left(-\frac{2GM}{r^2} \right) = \frac{GM}{r^2 \alpha} \quad (\text{from } \partial_r g_{tt})$$

$$\Rightarrow \Gamma_{tt}^r = \frac{1}{2} g^{rr} (-\partial_r g_{tt}) = \frac{1}{2} \alpha \left(\frac{2GM}{r^2} \right) = \frac{GM\alpha}{r^2}$$

$$\Rightarrow \Gamma_{rr}^r = \frac{1}{2} g^{rr} (\partial_r g_{rr}) = \frac{1}{2} \alpha \left(-\frac{2GM}{r^2 \alpha^2} \right) = -\frac{GM}{r^2 \alpha}$$

$$\Rightarrow \Gamma_{\theta\theta}^{\eta} = \frac{1}{2} g^{\eta\eta} (-\partial_{\eta} g_{\theta\theta}) = \frac{1}{2} \alpha (-2\eta)$$

$$\Rightarrow \Gamma_{\phi\phi}^{\eta} = \frac{1}{2} g^{\eta\eta} (-\partial_{\eta} g_{\phi\phi}) = \frac{1}{2} \alpha (-2\eta \sin^2 \theta)$$

$$\Rightarrow \Gamma_{\eta\theta}^{\theta} = \Gamma_{\theta\eta}^{\theta} = \frac{1}{\eta} \text{ (unchanged from Q.1)}$$

$$\Rightarrow \Gamma_{\phi\phi}^{\theta} = -\sin \theta \cos \theta \text{ (same as Q.1)}$$

$$\Rightarrow \Gamma_{\eta\phi}^{\phi} = \Gamma_{\phi\eta}^{\phi} = \frac{1}{\eta} \text{ (1)}$$

$$\Rightarrow \Gamma_{\theta\phi}^{\phi} = \cot \theta \text{ (1)}$$

Q.3

The Riemann Curvature tensor is $R_{\mu\nu}^{\rho} =$

$$\partial_{\mu} \Gamma_{\nu\sigma}^{\rho} - \partial_{\nu} \Gamma_{\mu\sigma}^{\rho} + \Gamma_{\mu\lambda}^{\rho} \Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho} \Gamma_{\mu\sigma}^{\lambda}$$

The Ricci tensor is $R_{\mu\nu} = R_{\mu\lambda\nu}^{\lambda} = g^{\lambda\sigma} R_{\sigma\mu\lambda\nu}$

In the Newtonian limit, the metric satisfies

$$g_{00} \approx -(1+2\Phi), \quad g_{0i} = 0, \quad g_{ij} \approx \delta_{ij}, \quad g^{\alpha\beta} \approx (-1, \delta^{ij})$$

$g^{ij} \approx \delta^{ij}$, with $|\Phi| \ll 1$, static fields ($\partial_0 = 0$) & velocities $\ll c$.

To leading order in Φ , the relevant Christoffel symbols are $\Gamma_{00}^i \approx \frac{1}{2} g^{ii} (-\partial_i g_{00}) \approx \frac{1}{2} (1) (-(-2\partial_i \Phi))$

& other Γ_{00}^λ are $O(\Phi^2)$ or zero. $= \partial_i \Phi$

& the Ricci component, $R_{00} = \partial_\lambda \Gamma_{00}^\lambda - \partial_0 \Gamma_{\lambda 0}^\lambda + \Gamma_{\lambda \mu}^\lambda \Gamma_{00}^\mu - \Gamma_{0\mu}^\lambda \Gamma_{\lambda 0}^\mu$

$\therefore$ Time derivatives ~~are~~, ∂_0 vanish & quadratic terms in Γ are $O(\Phi^2)$ and negligible; thus,

$$\boxed{R_{00} \approx \partial_i \Gamma_{00}^i = \partial_i (\partial^i \Phi) = \delta^{ij} \partial_i \partial_j \Phi = \nabla^2 \Phi}$$

(sum over spatial indices $i, j = 1, 2, 3$)

Q (4) Einstein field Eqⁿ.

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi G T_{\mu\nu}$$

Now, contracting with $g^{\mu\nu} \Rightarrow R^\mu_\nu - \frac{1}{2} R \delta^\mu_\nu = 8\pi G T^\mu_\nu$

$$\Rightarrow R - 2R = 8\pi G T \quad (\text{where } T = T^\mu_\mu = g^{\mu\nu} T_{\mu\nu})$$

$$\Rightarrow R = -8\pi G T$$

For a static, pressureless dust distribution, the energy-momentum tensor is $T^{\mu\nu} = \text{diag}(\rho_m, 0, 0, 0)$,

Where ρ_m is the mass-energy density and the 4-velocity $u^\mu = (1, 0, 0, 0)$.

$$\begin{aligned} \text{Lowering indices are } \Rightarrow T_{00} &= g_{0\alpha} g_{0\beta} T^{\alpha\beta} \\ &\approx g_{00}^2 T^{00} \\ &\approx (-1-2\phi)^2 \rho_m \\ &\approx \rho_m \end{aligned}$$

$$\begin{aligned} \text{The mixed tensor satisfies } T_0^0 &= g^{0\sigma} T_{\sigma 0} \\ &\approx g^{00} T_{00} \\ &\approx -(1-2\phi) \rho_m \\ &\approx -\rho_m, \end{aligned}$$

$$\text{and as noted, } T_0^0 = -T^{00} \approx -\rho_m$$

$$\text{The spatial components } T_j^i \approx 0$$

$$\text{The trace is } T = g^{\mu\nu} T_{\mu\nu} = g^{00} T_{00} + g^{ij} T_{ij} \approx g^{00} T_{00} +$$

$$0 \approx -(1)\rho_m$$

($\because$ spatial force is negligible in weak field).

$$\text{Thus, } \boxed{R = -8\pi G(-\rho_m) = 8\pi G \rho_m}$$

Q. (5) From Q. 3, in Newtonian limit, $R_{00} \approx \nabla^2 \phi$

Consider the $(0,0)$ -component of the Einstein field eqⁿs: $R_{00} - \frac{1}{2} R g_{00} = 8\pi G T_{00}$.

With $T_{00} \approx \rho_m$ and $g_{00} \approx -1$, this becomes

$$R_{00} + \frac{1}{2} R \approx 8\pi G \rho_m$$

From Q. 4 $\Rightarrow R \approx 8\pi G \rho_m$, so $\frac{1}{2} R \approx 4\pi G \rho_m$

Substituting into the $(0,0)$ -eqⁿ: $\nabla^2 \phi + 4\pi G \rho_m \approx 8\pi G \rho_m$

This simplifies to $\nabla^2 \phi = 4\pi G \rho_m$